Roger Torres

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Education

University of Pittsburgh — Pittsburgh, PA
Bachelor of Science in Computer Science — *Expected May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Algorithm Implementation, Linear Algebra, Computer Organization & Assembly Language, Mathematical Foundations of Machine Learning, Discrete Mathematics

Experience

Synopsys — Canonsburg, PA
Software Development Intern — *August 2026 – Present*

- Resolved a 14s UI freeze when right-clicking a selection of 2,000+ bodies, cutting the offending method from 82% to 0.12% of CPU time in profiling by adding an early-exit check in C# that stops at the first valid selection.
- Reduced duplicate scripting API surface in SpaceClaim/Discovery by removing 18 redundant overloads and deprecating 30 more across 24 files, using optional parameters and Obsolete wrappers so existing user scripts kept working.
- Automated adding new CoreService and DMS API versions with GitHub Copilot agents that reproduced 126 of 127 and 29 of 29 files from reference PRs in a single prompt, by testing against each reference, diffing the output, and rewriting the instructions until they matched.
- Fixed a multi-selection bug in SpaceClaim/Discovery where transformations applied only to the first selected face, edge, or body, by reworking C# selection methods to support targets across multiple components, verified with an 8-test journal.

University of Pittsburgh — Pittsburgh, PA
Undergraduate Teaching Assistant & Peer Tutor — *August 2026 – Present*

- Serve as UTA for Data Structures & Algorithms, leading office hours and grading for 100+ students.
- Tutor Computer Science students one-on-one and in small groups weekly, reinforcing core concepts across data structures, algorithms, and systems coursework.

Projects

Pulseboard | *Java 17, Spring Boot 3, React 18, TypeScript, PostgreSQL 14, JPA, JUnit 5*

- Built a productivity analytics dashboard with 16 REST endpoints via Spring Boot, tracking tasks and visualizing streaks, weekly totals, and completion rates.
- Modeled relational data with PostgreSQL and JPA; added JWT auth with user-scoped data access and sub-4ms query latency.
- Built deadline logic that flagged 85% of at-risk tasks via automated alerts before due dates, backed by 22 JUnit 5 tests.

Rehearse | *TypeScript, Next.js, React, Node.js, Express, WebSockets, OpenAI Realtime API, Judge0*

- Built a voice-driven mock technical interviewer where an AI interviewer talks through LeetCode-style problems in real time and sees the candidate's code and test results as they work.
- Built a Node/Express WebSocket server that syncs editor state to the interviewer and mints short-lived OpenAI Realtime tokens so API keys never reach the browser.
- Integrated Judge0 to run Python, JavaScript, Java, and C++ against per-language test harnesses on 15 original problems, and generated scored feedback reports with Claude, backed by 19 test files.

KeyGuide | *React, TypeScript, Tone.js, Vite, Tailwind CSS, Python*

- Built a browser-based piano learning app with a 3-octave virtual keyboard, 3 practice modes (auto-play, step-through, MIDI wait-for-correct), and a searchable song library with difficulty ratings.
- Integrated Web MIDI API for real keyboard input with live visual feedback and Tone.js PolySynth for low-latency audio synthesis across 36 keys.
- Built a Python CLI pipeline using mido to convert MIDI files into structured song JSON with auto-detected hand splits and heuristic fingering assignments.

RunIt Pittsburgh | *React, Vite, Firebase, Leaflet, Tailwind CSS*

- Built a web app connecting basketball players to pickup games in real-time, mapping 25+ courts across Pittsburgh with Firebase-backed live check-ins.
- Developed an interactive Leaflet map with real-time court markers and player counts; achieved sub-1.5s page loads with Vite and responsive design via Tailwind CSS.
- Placed 2nd in the CGI competition.

Technical Skills

Languages: Python, Java, C#, SQL, JavaScript, TypeScript, HTML, CSS
Frameworks & Libraries: Spring Boot, React, Node.js, Express, JUnit, Tailwind CSS, Tone.js
Databases: PostgreSQL, Firebase
Tools: Git, GitHub, Maven, Linux